Macromate

AI Project Proposal

Team Members

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1. Problem and Users

Recording Pakistani meals in a nutrition log takes several steps: naming the dish, finding suitable nutrition information, and calculating the amount eaten. Macromate will make the first step easier by suggesting a dish from a photograph. Its focus is the recognition of selected Pakistani foods, including visually similar dishes such as biryani and pulao.

The intended users are adults who track food for general fitness goals. This is an academic prototype. It will help users log meals, but its nutrition figures will be estimates rather than medical guidance.

2. Prototype Scope

The initial target is eight dish categories: chicken biryani, chicken pulao, chicken karahi, daal, nihari, haleem, samosa, and chapati. We will confirm the final list after checking image and nutrition data. Each uploaded photograph should show one main food item. The system will not claim to recognize every Pakistani dish or measure serving weight from the image.

The user will upload a photo, review the suggested dish, correct it if needed, and enter the serving size in grams. The app will estimate calories, protein, carbohydrates, and fat, then add the meal to a daily log. If the dish is unsupported or the suggestion is wrong, the user can choose it manually. Simple goal-related messages may use the daily totals; they will be rule-based.

3. How the AI Will Work

We will use CLIP, a pretrained model that compares an image with written descriptions. For each supported dish, we will write a consistent prompt, such as 'a photo of Pakistani chicken biryani'. CLIP will score how closely the uploaded image matches each prompt and return the top suggestions. The user will confirm the result before nutrition is calculated.

First, zero-shot CLIP: We will run the original model without changing its weights and record its accuracy on Pakistani food photographs. This gives us a baseline.

Then, fine-tuned CLIP: We will train a small part of CLIP's image encoder with labeled Pakistani food photographs. The text encoder will remain fixed. During training, we will adjust the image encoder so each photo scores higher for its correct dish prompt. The app will still use CLIP's image-to-text scores; there will be no separate classifier. We will compare this version with the original model and integrate the better-performing version.

4. Data and Preparation

We aim to collect about 60-100 varied, correctly labeled photographs per dish, depending on available data. We will record each image's dish label and source. Photos should vary in lighting, camera angle, plate, and recipe appearance. We will also create a small nutrition reference table with values per 100 grams and a documented source or standard recipe for each dish.

Python scripts will check missing or inconsistent labels, unreadable images, duplicate or near-duplicate photos, and unusual nutrition values. We will use consistent names and units, and apply CLIP's required image resizing and normalization. Initial data charts will show image counts by dish, missing data, and sample photos. We plan to separate the images into training, validation, and test sets before fine-tuning; photos from the final test set will never be used for training or prompt selection.

5. Evaluation Plan

We will test zero-shot CLIP and fine-tuned CLIP on the **same untouched test photos**. The main result will be top-1 accuracy: the share of photos for which the first suggested dish is correct. We will also report top-3 accuracy, accuracy for each dish, and a confusion matrix showing which dishes are mixed up. Biryani versus pulao will be examined closely.

Our work sequence is: collect and clean photos; measure zero-shot accuracy; fine-tune a small part of CLIP using training photos; compare both versions on the held-out test set; and connect the better version to the app. We will report actual results, including errors, rather than assume fine-tuning improves performance. We will also check the serving-size calculation with known examples and confirm that users can correct a wrong dish suggestion.

6. Nutrition and Limits

Nutrition will be calculated from the confirmed dish and the user's serving weight: estimated value = reference value per 100 grams x serving grams / 100. Recipe ingredients, oil, and cooking methods vary, so a photograph cannot reveal exact calories or macronutrients. The app will label these numbers as estimates.

The first version will handle one main dish per photo. Mixed plates, hidden ingredients, automatic portion estimation, and personalized medical advice are outside its scope. These limits keep the project manageable while allowing us to test the central AI question: whether fine-tuning CLIP on Pakistani food photos improves recognition over zero-shot CLIP.